

Colistin Breakpoints

Guidance document 2022

Background

Over the last few years, the colistin breakpoints have been under discussion. A recent publication in Clinical Infectious Diseases [1] highlighted some of the limitations of this drug:

- Less than 50% of patients with normal renal function achieve proper drug exposure, and it is associated with high risk of nephrotoxicity.
- Even in patients achieving adequate drug exposure, this level of exposure is still insufficient to achieve bacterial stasis in pneumonia
- Randomized and observational studies consistently demonstrate increased mortality for polymyxins compared with alternative agents, when polymyxins are used in monotherapy

Due to all these concerns, CLSI decided to report wild type microorganisms as “intermediate” to colistin. This would not be a good fit with the EUCAST S/I/R-definitions, as this “I-category” cannot be managed by increased drug exposure.

Recently, EUCAST has introduced brackets to deal with the situation where breakpoints have important caveats [2]. Essentially, the use of brackets means that the breakpoints can be used to distinguish between bacteria with and without acquired resistance mechanisms, but that the drug should not be used in monotherapy, unless used in infections where high exposure can be achieved at the site of infection (infections emanating from the urinary tract). For systemic infections, colistin must be used in combination with other active therapy. In this circumstance, the breakpoint in brackets (essentially an ECOFF) can be used to distinguish between organisms with and without acquired resistance mechanisms. For isolates without resistance mechanisms, include a comment in the report: “Colistin is often given in combination with other agents, either to support the activity of the agent or to broaden the spectrum of therapy. In systemic infections, colistin must be supported by other **active** therapy.”

Action

In breakpoint table v 12.0 (2022), in addition to those already introduced for aminoglycosides, EUCAST introduce brackets for all colistin breakpoints and for clindamycin and *Bacteroides* spp. The breakpoints will remain ECOFF-based

following these changes, but importantly reporting without communicating caveats should not be done.

Tab	Previous breakpoints		Breakpoint Tables 12.0	
	S ≤	R >	S ≤	R >
<i>Enterobacterales</i>	2	2	(2)	(2)
<i>Pseudomonas</i> spp.*	2	2	(4)	(4)
<i>Acinetobacter</i> spp.	2	2	(2)	(2)

References

1. Michael J. Satlin, James S. Lewis II, Melvin P. Weinstein, Jean Patel, and Romney M. Humphries (for CLSI); Gunnar Kahlmeter, Christian G. Giske, and John Turnidge (for EUCAST). Clin Inf Dis 2020; 71: e523-9
2. Guidance document EUCAST breakpoints in brackets 1 December, 2021.
https://www.eucast.org/fileadmin/src/media/PDFs/EUCAST_files/Guidance_documents/Breakpoints_in_brackets.pdf